

ML for cluster scheduling

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Scheduling

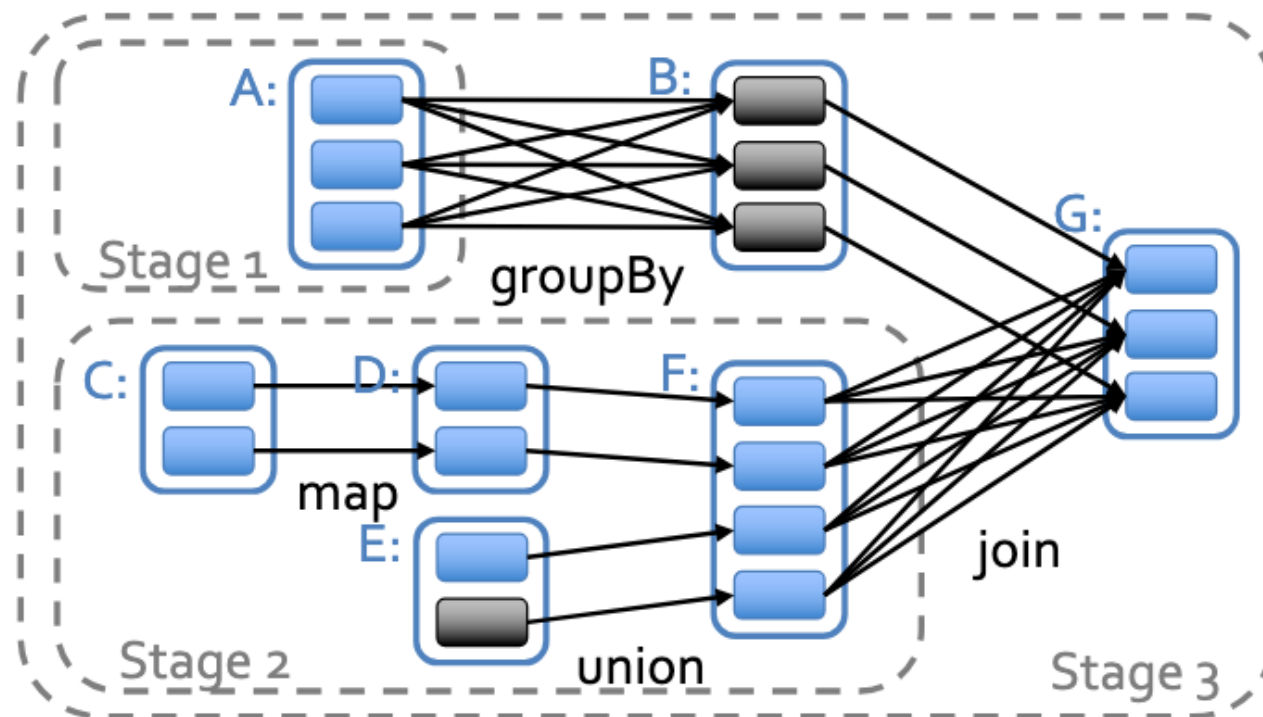
- Metrics and goals?
- Some common policies?

Spark

- Distributed computing framework
- Break task into smaller units/jobs
- Combine jobs
 - Map, filter, union, join, groupBy

Spark scheduling

- Cache-aware for data reuse, locality
- Partitioning-aware to avoid shuffles



Graph neural networks

- Learn the graph
 - Node features
 - Neighbors and their features
 - Distance between nodes
- Common examples:
 - Social network graph
 - Related papers (ConnectedPapers)

RL + policy gradient

- RL goal:
 - Pick action based on state
 - State-action space can sometimes be large
 - Use function approximators to map state-actions
 - Functions: can be as simple as linear combination of features
 - Functions: neural networks
- Training the neural network function: policy gradient

Discussion

- Would it be possible to train the GNN and policy network separately?
- How to optimize for other metrics?